Evaluation Report for [AccompTrainy]

[HXH, Tutorial #3]
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Introduction

HxH went into this project with a problem with the commuting experience at the forefront of our minds, with the main aim of the project attempting to minimise the travel anxiety and safety concerns that many commuters suffer from when travelling during the late hours of the night. This belief that commuters often felt uneasy when travelling late at night was only further developed through the various user interviews that the team conducted. With the ubiquitous negative emotions present within all interviewees revolving around late night commuting presented a clear issue with the commuter experience that HxH could tackle.

The team conducted 18 separate interviews with 18 unique participants, each with their own unique life experiences and expectations. To condense this large amount of user data our persona "Hanna" was created to encompass all our unique participants. (See Appendix A.) Hanna represents a young female university student, who is attempting to balance a postgraduate degree with work and keeping a healthy social life. She uses public transportation almost religiously when commuting to and from university, which with the large workload of a postgraduate degree often leads to her travelling alone during the late hours of the evening. While she is well accustomed to public transportation even after several years of late night travel she still feels uneasy, anxious and fears for her safety while travelling during the odd hours of the night. The persona also offers a 'face' for us as designers to specifically target aspects of the project towards.

Through the data collected by the interviews and the persona developed the main cause for the negative emotions and feelings during these late night commutes was discovered to be travelling in isolation. HxH's solution to this key problem in the late night commuting experience was the mobile application, 'Accomptrainy", which as an application focuses on bringing together like minded individuals in an attempt to create social engagement to combat this isolation. Accomptrainy aims to achieve this through the use of networking users to form travel 'pods' or groups to enable a safety in numbers effect.

Accomptrainy's paramount feature is its ability to facilitate user connectivity through the creation and joining of groups. This Level 1 functionality allows users to connect with others by selecting their departure place, destination, and departure time, ensuring they can coordinate travel plans. At Level 2, the app prioritises safety by providing report and emergency features, allowing users to report incidents or call for help in case of unsafe situations during travel. Additionally, at Level 3, users can manage connections with trusted individuals, referred to as buddies, and review past trips for future reference or security tracking. Together, these features ensure Accomptrainy offers both social connection and personal safety.

The Prototype

The Accomptrainy prototype used for both the usability and experience evaluations was constructed using the Figma application, a collaborative web-design tool regarded as an industry standard for UI/UX design. Accomptrainy is a mobile application created in hopes to enhance the everyday commuters, specifically university students commuting during odd hours of the day due to late university classes. The decision of opting for a mobile application rather than a desktop application stems from the convenience and accessibility it offers, as well as the responses gathered from the 18 interviewees, who highlighted a preference in using similar applications such as Tripview and Life360 for its ease of access on the go and its ability to update real-time tracking.

Using the Figma software, a high-fidelity prototype was able to be produced through variations of group ideations and collaboration. The prototype initially began from a storyboard, where each team member was given free reign in designing an individual frame of the wireframe. Placement of buttons and pictures was discussed thoroughly during the ideation sessions and soon enough, after completion, the low-fidelity prototype was produced. Once the lo-fi wireframe was created, each team member was tasked to conduct a heuristic evaluation based on the newly-developed prototype. The evaluation enabled the group to have a deeper understanding of the ecological relationship that revolves between the user's interaction with the application and the emotions the user may experience depending on the useability and overall design of the interface. After condensing everyone's heuristic evaluation, the group agreed to keep functions such as a back button on the bottom of the screen which allowed for ease of access with the thumb rather than a top screen backspace due to its inconvenience. The visuality of the application was another topic for discussion as the team believed having a captivating colour palette and a minimalistic aesthetic can benefit user experience, making the interface more engaging and visually appealing.

The creation of the mid-fidelity prototype incorporated all feedback derived from the heuristic evaluations, visuality concerns and most importantly, a functional logical flow. The application boasts a flat design, removing unnecessary styling and elements that may be seen as cluttered. The primary colour of the application is a dark neon blue, chosen for its inviting nature and its ability to ease the user’s eye, followed by a secondary colour palette of different shades of grey, enabling a minimalist and clean aesthetic. After a consistent pattern was established, a logical flow was implemented to replicate the user’s experience. A report system where the application will instantly call a transport officer when required was brought up by a team member during an ideation meeting to account for a potential user scenario where a user may experience a disturbance to another app user where immediate help would be required, a recall to recognition to application with this exact feature such as Life360 and Uber. The ability to remove and add people within the same lobby was another feature that was noticed during the Mid-Fidelity evaluations which was promptly added to the logical flow in order to give the user a level of control and freedom to the way they choose to use the application. Finally, after refining the Mid-Fidelity Prototype, each member was tasked to conduct a usability testing with a user that met the requirement of our target audience. This testing enabled the group to

gather an outside opinion as well as pickup any errors or mistakes recognized by the testing user. A High-Fidelity prototype was developed after refining the mid-fidelity version based on the testing user's feedback, ensuring the design more accurately aligned with user expectation and requirements.

The Evaluation

This report presents 2 assessment results of the "Accomptrainy" prototype to examine its useability and user experience. The examination is separated into two main parts, Part A and Component B, each of which concentrates on particular elements of the customer's communication with the prototype.

Part A (pages 7 to 26) includes a useability test report, which intends to figure out the useability advantages and disadvantages of the prototype with an extensive evaluation of user interaction. The major goal of this assessment is to determine just how very easy it is for individuals to finish key tasks, determine any possible obstacles or frustrations they face, and make specific suggestions for improvement to enhance the overall useability of the application. By concentrating on core functions such as itinerary creation, connection and security functions, Part An offers understandings into just how prototypes adhere to useability standards and principles, focussing on creating a safe and user-friendly user journey.

Component B (pages 27 to 32) reports the experience evaluation, which focuses on comprehending the individual's emotional response and their overall experience when interacting with the "Accomptrainy" application. The assessment intends to evaluate just how the application can effectively meet the requirements and assumptions of the target audience - specifically the safety and comfort when travelling throughout off-peak hours. The evaluation offers insight right into the success of the prototype in decreasing travel-related anxiety, developing trust among travel companions, and making certain that the application experience is positive and encouraging for individuals.

With each other, these analyses aim to provide a total understanding of the "Accomptrainy" prototype, while addressing functional aspects (such as ease of use and navigation efficiency) and experience factors (such as user satisfaction and emotional comfort). This thorough assessment technique permits a detailed review to make sure that the actual useability and the user's subjective experience are optimised for a much safer and extra interconnected trip late at night.

Executive Summary

The purpose of these evaluations was to assess how effectively the Accomptrainy app functions in real-world settings and meets user needs. The goal was to identify key issues in the mid-fidelity prototype and gather insights for improvements. Usability testing was conducted both online, via Zoom, and in person, allowing us to observe participants' initial interactions with the app. A scripted user scenario was created for the user experience evaluation to simulate a real-life situation, ensuring authentic feedback.

The evaluations provided a thorough analysis of the app's functionality across various scenarios, based on the user story. Conducted between October 8th and October 18th, 2024, the usability tests captured participant reactions in different environments. For the user experience evaluation, filmed on October 19th, 2024, a realistic scenario was staged, where a user navigated their trip home after a late class using the Accomptrainy app.

Through these tests, several issues were identified. Problems with consistency and alignment were apparent, including misaligned text boxes and titles, inconsistent button placements, and insufficient instructions for new users, leading to confusion during navigation. Key findings from the usability evaluation revealed mislabelled text, unstandardized buttons, and a non-intuitive method for returning to the dashboard.

- Misaligned text boxes and titles
- Inconsistent button placement across screens
- Lack of clear guidance for new users

From the user experience evaluation, our group identified the importance of efficiency and timely visual feedback to minimise user stress and improve overall interaction.

- Efficient travel planning and group creation features
- The need for fast, concise visual feedback to reduce stress and anxiety

Based on the identified issues, our team made several recommendations, such as standardising button placement across screens, providing clear visual feedback, and improving the flow for first-time users. These recommendations are aimed at improving both the usability and user experience of the app.

PART A. USABILITY Evaluation

Methodology

Evaluation Heuristics

Throughout the project's development lifecycle, the team conducted multiple usability evaluations to ensure the design's effectiveness and usability. These evaluations largely relied on employing Nielsen's Usability Heuristics to gauge the prototype at various developmental stages. Nielsen's heuristics were specifically chosen as a basis for these evaluations, over the other design heuristics, due to the Nielsen's principles being a long standing presence as an industry standard with it simple but robust principles that allows for efficient review and evaluation of prototypes at all stages of development

The first evaluations completed by HxH, was a heuristic evaluation conducted on our initial low-fidelity prototype. We chose to evaluate the low-fidelity prototype for this initial assessment as it allowed us to identify any heuristic violations, specifically of visual designs and major functions, before large amounts of developmental time and resources were consumed.

Following on from the heuristic evaluation, our prototype was further developed using the results from said evaluation. Once the next stage of the product was to a satisfactory state, a further usability evaluation was conducted. The reasoning behind this secondary evaluation was much the same as the first, attempting to locate any issues that violated Nielsen's heuristics, but was focused more heavily on functionality of the application.

Recruitment and procedure

With our two unique evaluations two different approaches were needed for post user recruitment and the procedure to each of them. For the first evaluation that assessed the heuristics of the low-fidelity prototype, we decided to conduct an in-house expert review, where each of the team members of HxH conducted a mock run through of each of our chosen scenarios. This test enabled us to go step by step through the application and identify violations of Nielsen's Usability heuristics. Results were noted down as each session was being completed and an in depth evaluation took place once the testing was done based on these results. The evaluations were conducted separately and later combined together to get a cohesive idea about any issues present while limiting the chance for violations to be missed.

The usability evaluation was conducted out of house, with each team member of HxH, finding a willing participant that fit within the scope of our target demographic, with all participants being university students aged between 20-25. By keeping participants within the target demographic it allows for any results found to be directly applied to the prototype after review. The evaluation was conducted through a number of steps, the first of which being a brief explanation of what the user is expected to do along with introducing them to the scenarios. By briefing them to the expectations it enables them to proceed through the other steps with little input from the tester. After the user has been introduced to the testing recording software step up, recording differs between testers with a combination of video recordings, screen recordings and voice recordings. With everything set up and ready, the user was instructed to go through each of the scenarios talking aloud as they went through the various screens. While they go through the application the tester recorded notes, focusing on their actions, words and body language. Once the evaluations were completed each individual team member reviewed their notes and recordings and catalogued the results for evaluation.

Evaluation Scenarios for Usability Evaluation

Scenario 1: Create a Trip

Goal: To assess how effectively users can create a new trip, manage group applications, and invite buddies.

- **Scenario Description**:
  - The user needs to create a trip leaving from **UTS at 8:30 PM** to **Hurstville**.
  - During the creation process, the user is required to set the **meeting spot for the group** at **Building 11**.
  - After creating the trip, the user should review the **applications** from other users who want to join the group.
  - Finally, the user should **invite one of their buddies** to join the trip.
- **Tasks Evaluated**:

1. Setting up a new trip with specific departure time and destination.

2. Setting a group meeting point.

3. Reviewing and accepting applications from others.

4. Inviting an existing buddy.

- **Heuristic Focus**:
  - **Match Between System and the Real World**: The user should understand how to set meeting points and times intuitively.
  - **User Control and Freedom**: The user should have flexibility to modify or cancel trip creation steps.
  - **Recognition Rather than Recall**: Reviewing and selecting buddies should involve minimal cognitive effort.
- **Expected Challenges**:
  - Users might struggle to locate or confirm the **meeting point**.
  - Reviewing and **inviting buddies** may lack clarity if the options are not clearly presented.

Scenario 2: Join a Trip and Contact Help in an Emergency

Goal: To evaluate how users join an existing trip and how effectively they can access emergency support features.

- **Scenario Description**:
  - The user needs to **join a trip** leaving from **UTS at 8:30 PM** to **Hurstville**.
  - The user will **view the first available group** and inspect the **members** of that group.

    - During the journey, the user encounters an incident that makes them feel **unsafe**, and they attempt to contact **help** using the emergency feature.
- **Tasks Evaluated**:

1. Joining an existing trip based on departure time and destination.

2. Viewing group members to evaluate the safety of joining the trip.

3. Using the **emergency button** or feature when feeling unsafe.

- **Heuristic Focus**:
    - **Visibility of System Status**: The user should clearly see their trip status and the members in the group.
    - **Error Prevention**: Ensuring that the **emergency button** is easily accessible and not prone to accidental presses.
    - **User Control and Freedom**: The user should be able to exit or seek help at any time.
- **Expected Challenges**:
    - Users might have difficulty identifying how to quickly view **group members**.
    - The **emergency button** may not be prominent or may not provide sufficient feedback when pressed, leading to confusion.

Scenario 3: View Past Trips and Report a User

Goal: To evaluate how effectively users can navigate their trip history and report problematic behaviour from a past trip.

- **Scenario Description**:
    - The user recalls that in their most recent past trip, **Johnny**, one of their trip mates, made them feel uncomfortable.
    - The user needs to **navigate to their past trips** and locate the trip involving Johnny.
    - The user then needs to **report Johnny** for his actions.
- **Tasks Evaluated**:

1. Navigating to **past trips** using the app's interface.

2. Locating the correct trip and identifying the specific member (Johnny).

3. Submitting a **report** detailing the issue with Johnny.

- **Heuristic Focus**:

    - **Recognition Rather than Recall**: The past trips should be easily accessible, and users should not have to remember details to navigate to them.
    - **Help and Documentation**: The reporting process should be clear, providing guidance to users on how to report and what happens afterward.
    - **Match Between System and the Real World**: The process of reporting should be intuitive and should match users' mental models of how such a feature would work.
- **Expected Challenges**:
    - Users may find it challenging to locate **past trips** if the navigation isn't straightforward.
    - The **reporting process** could lack clarity, especially if users are unsure of what happens after they report an issue.

Results

Following the 3 scenarios, the overall usability of the solution tested was highly positive, with most users finding the application interface to be intuitive and easy to navigate. The flat design, minimalistic colour palette and seamless logical flow allowed for a smooth user experience, mitigating confusion and cognitive load. The placement of the many buttons, photos and text boxes also contributed towards the completion of the scenarios. However, many inconsistencies to the formatting of text, titles, etc. were noticed by the users which definitely needs to be addressed in the high-fidelity prototype. Resolving these errors and refining the visual elements will further enhance user experience.

Major usability issues

Scenario 1: Create a trip

Task	Issue	Relevant Heuristic	Severity
What is the task	Describe the issue related to this task (one cell per issue)		
Navigate to create a trip	User goes to the "Create new trip" button on the dashboard but the button may not be clear to the user as it says both create and join new trips.	Recognition not recall- The unclear button at the top relies on the user's previous experience, which can confuse new users who may not know what action to take.	High
Selecting a specific starting point	Users have difficulty selecting a starting location quickly as the screen lacks direct	Flexibility/ efficiency of use - The screen is inefficient, as it lacks an option for users	Medium

	navigation for frequent locations.	who are familiar with the app and know their desired location to quickly navigate to it directly.	
Selecting a specific destination	Similar to the starting point, selecting an end location lacks options for direct navigation for frequently used destinations.	Flexibility/ efficiency of use - The screen is inefficient, as it lacks an option for users who are familiar with the app and know their desired location to quickly navigate to it directly.	Medium
Selecting a trip time	The trip time selection process is inefficient and requires multiple unnecessary steps.	Flexibility/ efficiency of use - The screen is inefficient, as it lacks an option for users who are familiar with the app and know their desired location to quickly navigate to it directly.	Medium
Creating a new group	The app lacks visual indicators for users with disabilities, making it difficult for them to navigate without additional support.	Aesthetic/minimal design - Individuals with disabilities, such as blindness, or those who are primarily visual learners, may struggle to understand what is happening due to the lack of visual indicators or cues.	High
Confirmation on meeting spot	There is no system feedback confirming the starting meeting point, leaving users unsure if the trip has been set up correctly.	Visibility of system status & feedback - The violation occurs because the system fails to display relevant information that keeps the user	Medium

		informed. The absence of specific stop details prevents users from knowing where they are on the route, making it difficult to track progress or plan their next actions, such as where to exit the route. This lack of visibility breaks the flow of interaction and undermines user confidence in navigating the app.	
Selecting applicants	Titles and buttons are inconsistently placed, causing difficulty when refreshing and reviewing applicants.	Consistency & standards - The misaligned title and unusual placement of the refresh button deviate from familiar design conventions. Users expect centred titles and easily accessible refresh buttons, so breaking these norms can cause confusion and hinder navigation. Aesthetic/minimal design - The misalignment and awkward button placement disrupt the app's visual simplicity. A clean, organised layout is essential for usability, and these issues create clutter, making the interface harder to navigate and less visually appealing.	Medium

Refreshing applicants	Only two applicants are visible at a time, and users cannot view more without returning to a previous step.	Flexibility/ efficiency of use - By only showing two applications and not offering a "show more" button, users are limited in their ability to explore more options. This adds unnecessary steps and reduces the flexibility of the interface for those looking to pair with more strangers.	Medium
Inviting a buddy to the group	There is no option to remove a buddy after adding them to the group, which limits user control.	User sense of control & freedom - There's no way to remove someone from a group after they've been invited, which restricts the user's ability to manage the group effectively, making it difficult to control the group's members and fix accidental invites.	High

Scenario 2 Join a trip and using emergency report button

Task	Issue	Relevant Heuristic	Severity
What is the task	Describe the issue related to this task (one cell per issue)		
Navigate to Join Trip	The button "New Trip" on the dashboard page is not clear to its function of being used to both join and create a group	Violates Recognition not recall, The buttons ambiguous label requires the user to learn the function of the button through experience.	Severe
Selecting time of Trip	The selection of the time for the trip to depart is present in a list of times.	User control and freedom: By forcing users to navigate through a list of available times rather	Mid

		than setting a chosen time severally limits their ability to control their trip planning	
Swiping to join a group page	The indication that the join group page can be found by swiping the application is minor and easy to miss	Visibility of system status and feedback: With the join trip prompt being hidden on a separate screen that is only accessible by either clicking a small button or swiping the screen with only a small dot as indication that it is even possible to do so, makes it extremely likely that users may miss that this feature even exists	Severe
Viewing groups	All groups are presented in one large group with no differentiation between public trips and the trips of “buddies”	Aesthetic / minimal design: With all of the groups bunched into one list it makes this page look cluttered and unordered. Recognition not recall: With no distinct difference between groups of “buddies” and public groups users are forced to remember who their buddies and manually look through each group to find their “buddies” group	Mid
Viewing groups / Joining a group	The crown symbol that is present next to some of the users in the viewing of groups page and group page is unclear to what it means	Match system to the real world: Due to the confusion expressed by various users the crown symbol was an incorrect choice to	Low

		represent a group leader Help and documentation: The lack of any tooltip that offers guidance to the meaning of the crown only further exemplifies user confusion.	
Joining a group	The only way to join a group is to go through the entire process with no way for a direct invitation to be received or accepted.	User sense of control and freedom: By forcing users to go through the entire process every time it limits the users ability to join groups through other ways.	High
Exiting group	No exit or leave group button present once the user has joined a group	User sense of control and freedom: The user is forced to stay in a group once they have joined. Help users with errors and recovery: The lack of a exit or leave group button is also a problem if the user mistakenly joined a group, and not allowing them to leave is a severe breach of this principle	Severe
Reporting a user	The report screens only have the functionality to show 2 users.	User sense of control and freedom: Does not allow for more than two people to be selected for reporting, limiting user options	Mid
Reporting a user	No way to leave report screen or report another user easily	User sense of control and freedom: Limiting options for how a user can go	Mid

		about reporting people breaches this principle. Error Prevention: While not having a cancel button readily available makes it hard for users to exit this screen if they did not intend to report someone.	

Scenario 3: View past trips and report user

Task	Issue	Relevant Heuristic	Severity
What is the task	Describe the issue related to this task (one cell per issue)		
View past trips	The user may want to interact with the trip details or icons, but it's unclear which elements are interactive such as the star element.	Visibility: The star icon for marking favourites lacks a clear indication of whether it has been successfully marked	Low
Viewing a past trip	The second screen for viewing trip details provides minimal information and doesn't indicate that group member profiles can be clicked for further details. The overall layout is sparse, without clear distinctions between different sections of the trip or interactive elements.	Recognition/recall: The screen forces the user to remember that clicking on a person's name will bring up more details rather than making it obvious through visual cues, such as buttons or icons.	Low
Report screen-initial selection	The "Help!" button stands out more than the report options. It is great for an emergency but it	Match between system and the real world: The red "Help!" button may imply emergency help, screen context is	Mid

	can confuse users about the purpose of the screen.	mainly for reporting though. Visibility of system status: It's unclear if the user knows they can select more than one person, or just one.	
User Selected	Select a specific user (Johnny Mane) to report. The selection of the user is clear, but there's no clear next step instruction to guide the user after the selection.	Recognition-recall: The layout is simple and easy to navigate through during moments of distress. User control and freedom: There should be a clear instruction or indication of the next step after selecting a user. But enabling users to choose and ensure that they selected the right user to report allows users to precise their selection and not the trip itself.	Low
Writing Report	The lack of a prompt or example text leaves the user unsure of how to describe the issue they are reporting.	Error prevention: Without guidance, users may input vague or insufficient information, potentially leading to issues with processing their report. Flexible/efficiency of use: Providing preset options or example text would speed up the process for the user and reduce cognitive load.	Mid

Confirm Report	The language used suggests that the user being reported will be banned, which may mislead users into thinking that reporting automatically results in a ban. The pop up window is also quite clustered.	Error prevention: The unclear wording may lead users to believe the problem users would be banned a review process with no immediate action Recognition not recall: The confirmation text doesn't make the process transparent. Users are left guessing how the report is handled immediately after submission rather than after review	Low
Report confirmation page	The message acknowledges the user's action and the actions it will take for their safety. There isn't availability of more options or actions to take afterwards	Visibility of system status & feedback: The user is reassured that their input was received, reinforcing trust in the app and encouraging further use if needed. It provides them with clear feedback that their report was successfully submitted and that further steps will follow. User sense of control & freedom: There is only one option to return to the dashboard, which is alright on a basic level but can be problematic in a developing situation.	High

Calling the emergency screen.	It could be better placed such as on the dashboard for easier access in an emergency. It could mean less clicking. Potential to expand variety to police or station rather than 000 right away. Misuse of services.	"Match system to real world: Replicates the simplistic design on many phones while users speed dial emergency services with an option to cancel Sense of control and freedom: The screen has a button for users to cancel the call if it is made recklessly or the issue has been resolved. However, this button can be easily misused upon the perpetrator gaining access to the phone in rare cases, impacting safety and purpose of the service. Recognition-recall: It requires users to remember that there is a panic button on the confirmation page, requiring learning how to access it rather than instantly access it. Consistency and standards: The design is minimalistic to avoid complexity during emergency situations but the design could have matched the brand better rather than sticking to basics.	Mid

Recommendations

Change	Justification	Severity
Specific instructions for change (one per cell)	Based on qualitative results, why is this change recommended	
Emergency button/report to be added to main dashboard page	This is necessary as the previous fidelity prototypes have this feature only accessible via confirmation, leaving a loophole where it wouldn't be accessible if there wasn't a way to access it on a dashboard setting	High
Implementation of sorting/filtering choices on create a trip	Similar applications and official apps implement this for their page rather than just search. A search function fulfils lookup functionality well but this experience can be elevated with sorting as users are made aware of nearest stations to them via lookup or map plugins.	Low
Implementation of scrolling date and time	Some users prefer the speed and ease of selecting from a scrolling picker, while others may prefer manually typing in an exact date or time for accuracy. Offering both options accommodates these preferences and supports users with different input habits.	Mid
The ability to create and join trips well in advance.	Many users, such as students attending events or repetitive classes, plan their journeys well in advance. Allowing trips and groups to be scheduled far ahead supports long-term organisation and allows users to plan ahead safely.	Mid
Feature to report users long after the trip via past trips and user page access.	Users may feel more comfortable or confident in reporting issues after they've had time to reflect on their experience, consult with others, or after realising that the behaviour was inappropriate. This feature supports reporting based on this delayed recognition.	High

Option to call station master of the next station, train operator or the police instead of direct calling to one on the calling emergency services page	Sometimes, situations can be resolved more peacefully by notifying the station master or train operator first, rather than escalating to police or emergency services. The presence of train staff may help calm a disorderly passenger without involving law enforcement, offering a more controlled approach. Providing multiple points of contact also gives users a greater sense of security, especially during late-night journeys when they may feel more vulnerable.	Mid
Add a "Show More" Button to Expand Applicant Lists	Users only see a limited number of applicants and have no option to view more unless they navigate back or use alternative actions. Adding a "Show More" button will provide flexibility in viewing the full list, aligning with the heuristic "Flexibility and efficiency of use." This change reduces user frustration during group management.	Med
Allow Users to Remove Buddies from a Group After Adding	Currently, once a buddy is added, there is no way to remove them from the group, which restricts user control. Providing an option to remove members will give users more flexibility and the ability to rectify mistakes, in alignment with "User control and freedom."	Severe
Make the Emergency Button Available on the Main Dashboard for Quick Access	During testing, users found it challenging to locate the emergency button quickly during stressful situations. By placing the emergency button prominently on the main dashboard, users can easily access help when needed. This addresses "Error prevention" and "User control," enhancing user safety during emergencies.	High
Emphasise Reporting Options on the Report Page, Instead of the Red "Help!" Button	On the reporting page, users were confused by the prominence of the red "Help!" button, as it detracts from the main purpose of reporting an incident. The visual hierarchy needs adjustment to prioritise reporting actions, which aligns with the heuristic "Match between system and real-world."	Med
Improve the Visual Feedback of Interactive Elements, Like the "Star" for Marking Favourites	Users did not always realise if the "Star" icon for marking favourites had successfully been clicked, due to insufficient feedback. Adding visual changes like colour changes or confirmation icons ensures users know their action has been acknowledged, aligning with "Visibility of system status."	Low

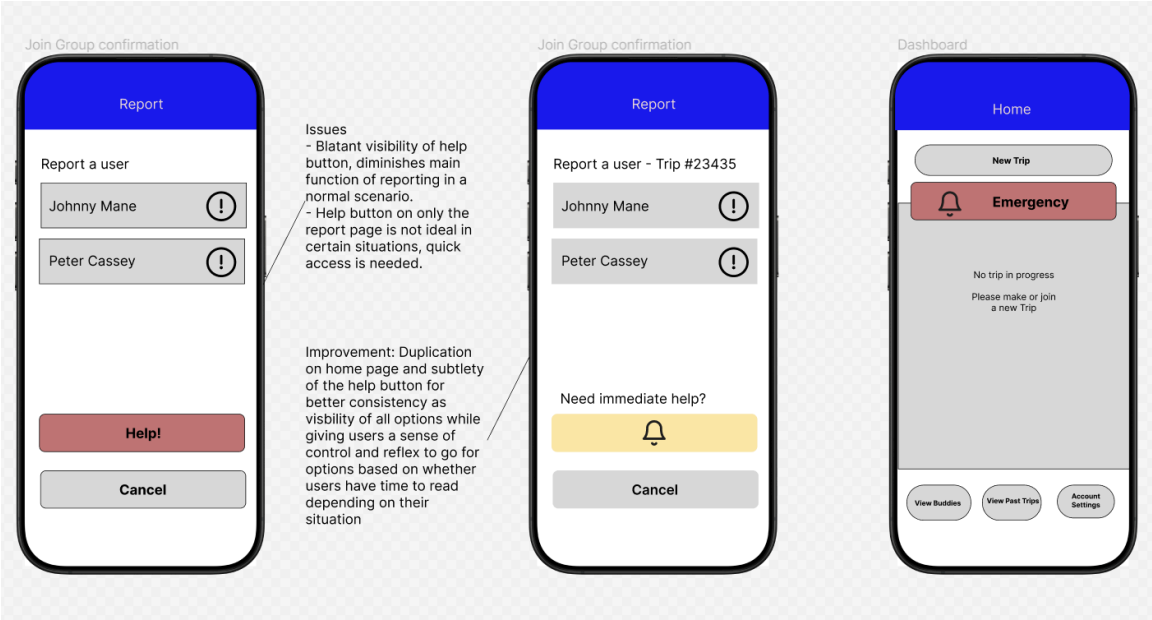
Create a Step-by-Step Onboarding Tutorial for First-Time Users	New users struggled to understand how to create or join trips, especially during their first interaction with the app. A step-by-step tutorial would guide new users and make it easier to learn the interface, aligning with the heuristic principle of "Help and documentation."	Med
Refine Visual Consistency Across All Pages by Standardising Button Locations	During testing, inconsistency in button locations across different screens led to confusion. Standardising button placement across all screens will make the navigation more predictable and intuitive, aligning with "Consistency and standards."	Med
Add a more visible button to return to the dashboard	The current navigation back to the dashboard requires the use of a back button, which is not always intuitive. Adding a dedicated dashboard button on every page will help streamline navigation and improve user flow.	Med
Redesign group joining interface to distinguish between public and "buddies" trips	Currently, the trips are presented in a single list without any differentiation. This makes it confusing for users who want to distinguish between joining a public group or a trip with friends. Adding visual cues or categories can enhance usability.	Med
Enhance the "Help!" button's context clarity	The emergency "Help!" button is placed alongside report options, which could confuse users about its use. Providing a tooltip or description to clarify that this is for emergency contact only will make its purpose clearer.	Med
Provide clear next-step guidance after selecting a user to report	After selecting a user to report, there is no explicit instruction on what to do next. Adding an on-screen prompt to guide users to the next action will prevent confusion.	Low
Include preset options or example text for reporting incidents	Currently, users must write reports without guidance, which may result in vague descriptions. Offering example templates or preset text will assist users in providing useful information, thus preventing errors during the reporting process.	Med
Add options after reporting a user, such as blocking or muting	After submitting a report, users are only returned to the dashboard without further actions available. Including additional actions, like blocking the reported user or muting future interactions, gives users a greater sense of control over their safety.	High
Improve Button Labels Consistency	Labels like "Create New Trip" and "Join New Trip" can confuse users due to the combination of both functions in one button. It's recommended to	Severe

	split these into distinct actions, e.g., "Create Trip" and "Join Trip" for clarity and to reduce confusion.	
Introduce Visual Feedback for Successful Actions	Users require better visual confirmation when selecting certain elements, such as marking a trip as a favourite or successfully joining a group. Adding a clear checkmark or change in colour can help with better visibility and reduce uncertainty.	Med
Enhance Error Prevention for Reporting	The current "Report" button's wording and placement can mislead users into thinking immediate action, such as banning, will occur. Clarify that reports will be reviewed first, and add brief documentation on the process to avoid confusion.	Low
Add Cancel Option for Invited Members	Users should be able to cancel or modify invitations sent to members. This provides users with better control and the ability to make corrections if they made a mistake while inviting someone.	Severe
Add Filtering Options to View Groups	Currently, there is no distinction between public trips and trips created by buddies, which makes it difficult for users to identify which type of trip to join. Adding filtering or categorization can make group selection faster and more intuitive.	Med

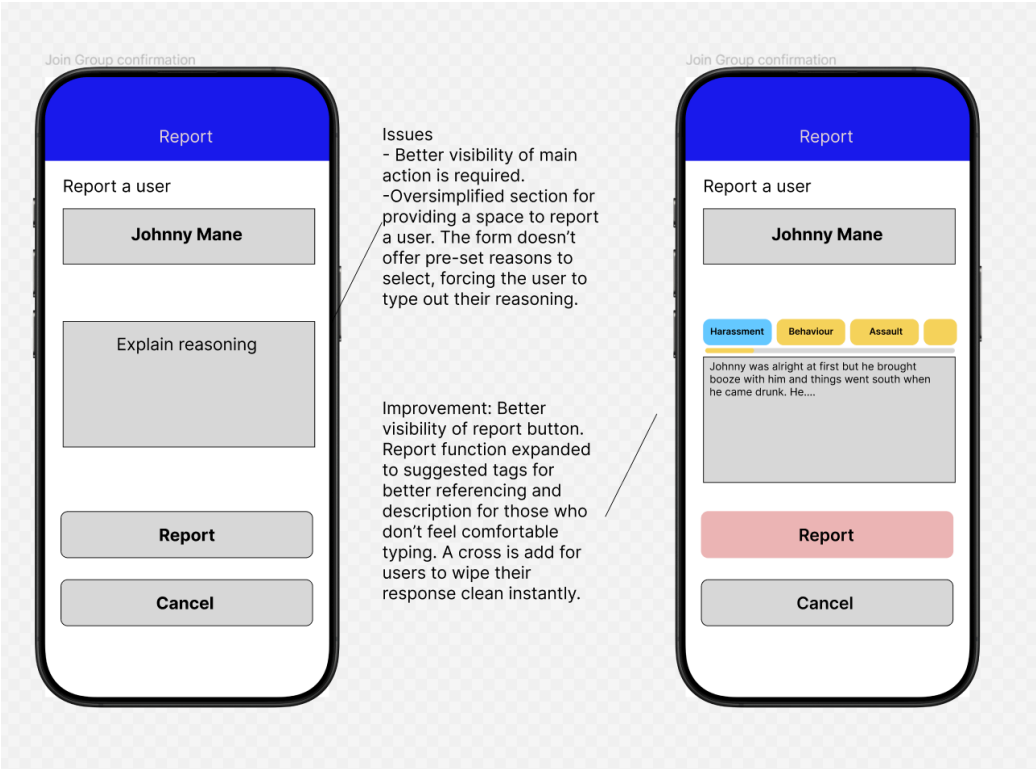
Improvements of prototype based on recommendations

Examples of improvements

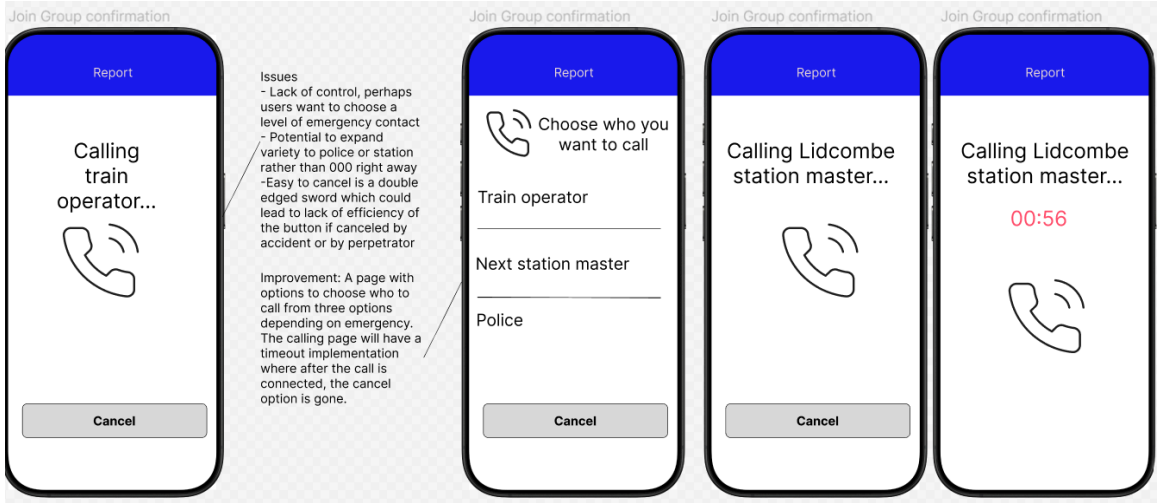
Screen 2: Report function



Screen 4: Writing report



Screen 7: Emergency calling



PART B. USER-EXPERIENCE EVALUATION

Methodology

Evaluation Approach

User-Experience evaluations were conducted in person on the UTS campus, where team members and users were instructed to follow a storyboard whilst they interacted with the Figma prototype in real-time. Throughout the recording, the user was tasked to complete two scenarios, which involved utilising all features of the application through specific, guided tasks as well as providing feedback through the perspective of an outside user on the application's interface, functionality of buttons, uniformity of texts and titles and the overall experience.

Procedure and Recruitment

Users chosen to participate in the evaluation included university students that typically take classes that occur during odd hours of the night, and utilise public transport as means of travel after said classes. A target demographic was introduced to ensure the application will be able to meet the needs and preferences of its intended users, of young adults that are exposed and vulnerable to potential threats during late night commute. The evaluation sessions were conducted in-person, involving an observer and a user, where the observer's role was to record the users every move and their reasonings using the heuristic evaluation sheet. The user was instructed to follow 2 scenarios which involved utilising all functionality found on the high-fidelity logical flow. The evaluation was not limited to a particular time limit as each scene of the chosen story board was to be recorded in real time with each task of the scenario prompt. Recordings were conducted on an iPhone 13, where the camera application was used to capture the entire process including user interactions, displaying the user's decision making on the Figma prototype, full body recordings, to show the users facials expression and emotions whilst attempting to complete the task. Additionally, the users' audio was recorded via the iPhone's inbuilt microphone, which allowed for the analysis of verbal feedback and any potential behavioural cues that may provide further insight to their experience with the application.

Evaluation Stories for Experience Evaluation

User experience stories were written to breathe life into largely data-driven persona, Hanna, and present realistic scenarios that users might encounter. They allow us as designers to understand how users could potentially interact with the application while simultaneously offering scenarios for user role playing for testing role play. The ability of these stories to humanise persona's is a key reason for why HxH endeavoured to create a pair of stories, each covering a principal aspect of Accomptrainy's main functionality.

The first of the user experience stories follows Hanna as she prepares to travel home late at night after a long exhausting day at university. Feeling anxious about travelling alone in the dark, she uses the Accomptrainy app to create a group trip for added safety. The main function highlighted by this story is creating a group, while also presenting the features that help facilitate this function.

The second of the stories follows a similar direction as the first with Hanna once again looking to travel home during the late night. However, this time, she opts to join an existing group instead of creating one. This highlights another key function of the app - joining an already established group. As she navigates through the app, she uses features such as viewing groups, group members, selecting her destination and departure time.

User story overviews (See Appendix Fig D, for complete user experience stories)

1. Hanna's emotional state is characterised by fatigue after a full day of classes and increasing anxiety about travelling at night. This story evaluates how effectively the app's features address these concerns, helping Hanna feel more secure.
2. Navigating through Hanna's feelings of fatigue and anxiousness as she joins a group, which later develops into fear when someone becomes threatening. The story addresses how users may feel before, during and after an uncomfortable situation and how the app helps users manage these feelings.

Results

In order to retrieve accurate results for our testing, we had decided to record a video where notes were taken and the user interaction with our subject was observed with their hesitation, words and expressions as well as the speed to get through the scenarios. Upon recording the video, the footage was then reviewed start to finish by the group who took notes on any noteworthy interactions that occurred

After the video had been thoroughly analysed the group then extrapolated the core functions that needed to be worked on, the factors involved in their usability and what recommendations could be implemented to make it better. Just because a task was achievable does not mean it was exempt from the possibility of being improved. This involved discussion amongst the group on what was the best approach we could recommend.

This information was then organised into tables with an analysis based on each feature, whether they violated or complied with the specific heuristic from Nielson, the reasoning behind it and recommendations to improve on negatives of each screen.

Create a Group

Phases	Main user experiences	Major factor affecting the UX	Recommendations
What the specific task was or goal	What was the user experience? *One user experience reported per row*	Describe the major factor impacting the UX and explain how this occurred	
Creating a group for travel safety	Hanna found the app interface intuitive, with clear buttons that helped her quickly create a group.	Positives: The usability of the app was strong, with simple navigation and an intuitive interface that helped Hanna set up a group easily. Negatives:Limited customisation options or more detailed group settings may limit more experienced users.	Keep the interface simple but allow for group customisation (e.g., preferred group size, security levels). Introduce a tutorial for new users.
Finalising the trip settings and forming a group	Hanna felt relieved that she was able to confirm the trip easily and meet her group members without any issues.	Positives: Efficiency was a key factor here, as Hanna could easily complete her task with minimal steps. The straightforward layout made the process quick and hassle-free.	Add a group chat feature or message prompts so users can interact and confirm details with group members directly before the trip starts.

		Negatives: Limited group interaction features: There was no built-in chat or group communication system to confirm or communicate with group members beforehand, which could cause confusion or uncertainty when waiting for others to show up.	
Waiting for group members after leaving university and travelling with them	Hanna was reassured that her group arrived promptly, which reinforced her positive perception of the app, wished that there was arrival notifications to prevent delays	Positives: The timely arrival of group members enhanced the feeling of reliability and made the experience smoother. The assurance that plans went as scheduled improved Hanna's trust in the app. Negatives: Lack of proactive status updates or notifications can leave users uncertain if group members are running late or on their way, potentially leading to unnecessary waiting time or anxiety.	Introduce an ETA system or proximity alerts to keep users informed about group members' estimated arrival times, helping to alleviate uncertainty and improve coordination.
Post-trip reflection	Hanna reflected that the trip was smooth, and the app provided a positive experience overall.	Positives: The overall user satisfaction with the app's core function was high, due to its ability to coordinate safe, group travel. Negatives: There was no follow-up or feedback system to report on the group experience or flag safety concerns after the trip.	Add a feedback feature where users can rate or report group members to ensure better future group matches.

Joining a group

Phases	Main user experiences	Major factor affecting the UX	Recommendations
What the specific task was or goal	What was the user experience? *One user experience reported per row*	Describe the major factor impacting the UX and explain how this occurred	
Joining a group and waiting for group members after leaving university.	Hanna felt reassured that her group members arrived on time, which reinforced her confidence in the app. However, the lack of real-time notifications or tracking made the waiting period slightly nerve-wracking.	Positive: The app's basic functionality of allowing users to form groups worked smoothly. The timeliness of the group members was a key positive aspect, ensuring that Hanna felt comfortable. Negative: A lack of real-time notifications or tracking when waiting for group members caused minor anxiety. Without knowing exactly when others would arrive, users might feel uncertain or unsafe in unfamiliar situations.	Implement real-time notifications or a live tracking feature to update users when their group members are nearby. This would reduce uncertainty and anxiety while waiting, especially during late-night travels.
Contacting help	Hanna felt safe being in a group but felt uneasy around one member of her group. This feeling of uneasiness turns into dread as she is required to get emergency assistance, she feels nervous waiting for this to work but feels slightly relieved when it's been received.	Positive: Having the emergency button close by mid-trip ensures it's immediately accessible. Giving people the confidence that they can immediately get help if things go wrong is necessary. Negative: Though it is meant to contact emergency services and on-site security in reality this app will probably be worse than actually just calling 000 normally. Since most times the more steps required only increases the chance of failure. In this case it's redundancy compared to existing solutions and	Implementing a redirect from the app to your phone app with the number pre-dialled would be an alternative alongside having a button to alert transport on-site security. It would give people greater confidence and prevent accidental fumblings in consideration for the panic people would be feeling.

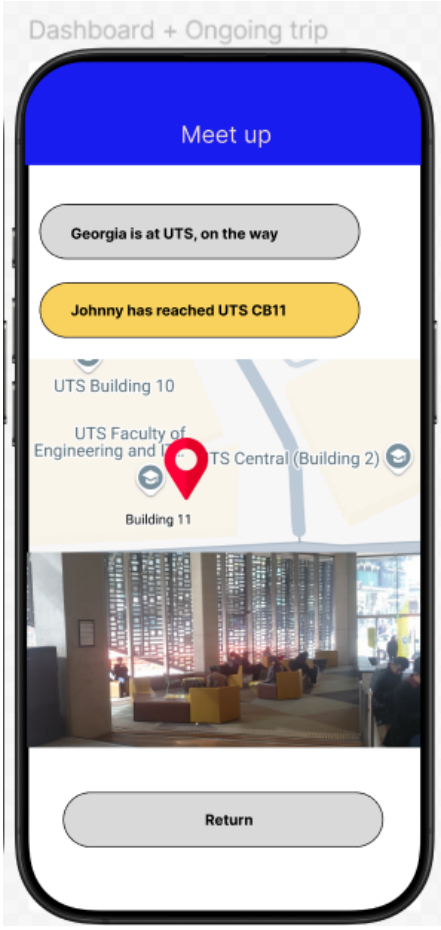
		options makes it questionable, such as emergency tram buttons.	

Improvements of prototype based on user experience evaluation recommendations

Meeting Spot Map

Image of meeting spot / finding group may be complicated with needing precise coordinates. A map with clear markers indicating specific meeting points or areas of focus. Including information about the event's purpose, schedule, and any necessary supplies would be beneficial to getting precise information. To waterproof this, a posted image of someone who's arrived at the meeting spot can send their current location. This can help tremendously with users finding the exact location via recognising common elements around them.

In a revision screen prototype, we have explored this option with a new screen.

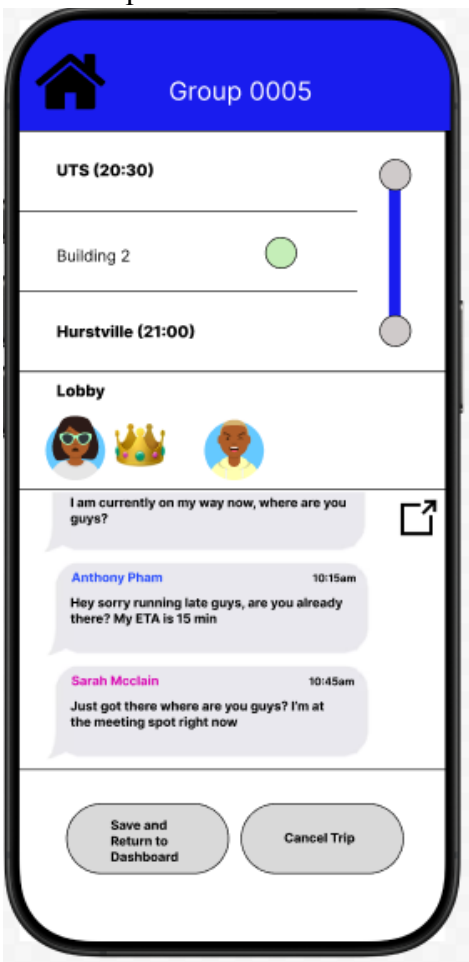


Group messaging

Upon reviewing the recommendations the group realised that further functions that would help with communication between members in a joined group would be required. As simple text based messaging can clarify a lot of queries between members and unlike image based communication requires less internet. As such texting is an essential component to any app involving people.

For managing meet ups and to inquire about any problems or occurrences that may affect the trip, see 'Join Group Travel Screen' below. Text messaging will have an option to enlarge to full-screen to view the entire text log easier and can then be minimised. This is available upon joining a group and will be made unavailable upon leaving the group. A text history will be available upon reviewing past trips.

Join Group Travel Screen



Limitations

While all of the different evaluations conducted provided valuable insights, each method came with their own unique limitations that may have impacted the results to varying degrees. Understanding these limitations is crucial for interpreting the findings.

The heuristic evaluation was conducted by team members, which introduced inherent bias into the process. With the evaluators being familiar with the design and intentions that each unique screen was portraying, overlooking certain usability issues or failing to assess various areas that an external reviewer might have found. Additionally, because of the low-fidelity nature of the prototype some usability errors may have been overlooked due to the limitation of the prototype, specifically visual elements might have been more apparent in a higher fidelity prototype, which may have caused some items to be overlooked.

The usability evaluation was conducted in a controlled environment, which may not have completely reflected real-world use cases. While the controlled setting allowed for careful observation of participants, it resulted in isolated results that might not translate well to actual environments where users may interact or use the application, such as noisy train stations or poorly lit areas. The absence of external distractions and real-world situations limits the ability for the evaluation to gauge the applications usability during a typical late night commute.

The user experience evaluation relied on team members roleplaying as the persona within the bounds of the user experience stories. This method, while incredibly useful for allowing team members to conceptualise a user's experience with using the application, introduced bias as the team members conducting the roleplay were intimately familiar with the app's design and intentions. This deep understanding of the app's supposed function may have skewed the results, with team members potentially overlooking design flaws or being less likely to encounter confusion compared to a real user. This bias could lead to overly optimistic evaluation when considering real-world use cases. By acknowledging these limitations, team HxH can better understand potential gaps in the evaluation process and leverage this knowledge for future assessments.

Conclusion

This report focused on a comprehensive evaluation of the Accomptrainy prototype, a mobile application designed to reduce travel anxiety and safety concerns for young adults commuting during odd hours of the night via a buddy system. Through rigorous usability testing on the basis of our persona, Hanna who encompassed our target market of young adults travelling at late night, employing Nielsen's Usability Heuristics, and user-experience evaluations based on user stories, the evaluation process has identified key areas of improvement for the application.

Usability testing revealed several areas where the prototype's interface could be made more intuitive and efficient. Issues such as unclear button labels, inefficient navigation, and lack of clear feedback were identified and addressed through specific recommendations for improvement. User-experience evaluations further highlighted the importance of clear communication, ease of access to emergency features, and a seamless reporting process in ensuring a positive and reassuring user experience.

The insights gained from both evaluations have been translated into actionable recommendations and implemented in the prototype. By addressing these usability and user-experience concerns, Accomptrainy can better serve its intended purpose of fostering a sense of safety and community among young adults commuting late at night.

Despite the controlled environment of the evaluations, the findings provide valuable insights into the effectiveness of the design and highlight areas for further improvement. The iterative nature of the design process, coupled with the feedback gathered from these evaluations, will contribute to the development of a user-centred application that effectively addresses the needs and concerns of its target audience.

Future iterations of Accomptrainy can benefit from further testing in real-world scenarios to assess its performance in diverse contexts and user populations. By continuing to prioritise user feedback and iterate on the design, Accomptrainy can evolve into a robust and reliable tool that empowers young adults to navigate their late-night commutes with confidence and peace of mind for high-fidelity and implementation stages of the application for users like Hanna to have a better experience while travelling on public transport safely.

PART C. APPENDIX

Appendix A. Persona

HANNA SMITH

Age: 22

Australian

Female

Caringbah, Sydney

Education

UTS

Masters of Physiotherapy

Bio

Hanna is a post grad student studying at UTS to complete her masters. She enjoys doing anything outside, while she tries to balance her work with her studies while keeping a healthy social life and a steady relationship.

Hobbies

Commuting

Hanna primarily relies on two types of transportation. For her daily commute to work or travelling locally, she predominantly uses her personal car. However, when traveling to university or heading out to Sydney, she opts for the train.

Apps

Concerns

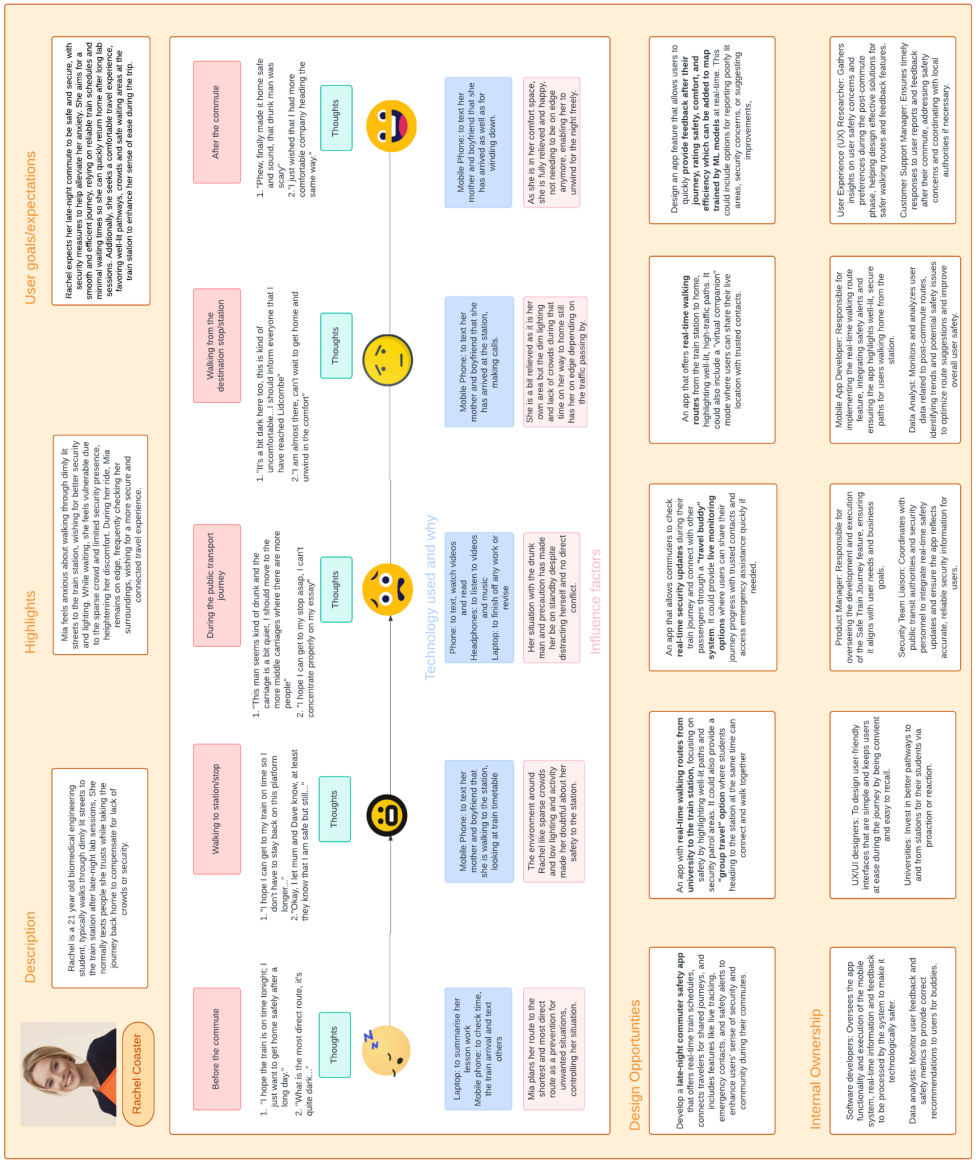
- Loud people and / or groups
- Large crowds
- Individuals under the influence
- Unpredictable people

Appendix B. Interview Recordings and Consents

Interviews

Appendix C. Journey Map

Journey Map



Appendix D. User experience stories x2

Fig D.1. User Experience Story 1: Create a Group

Hanna feels tired after long days at university and nervous about travelling during the night.
It's 9pm on a Friday night, and after 3 separate classes totaling 8 hours of total time spent at university, Hanna is ready to head home. It is now quite dark outside, and Hanna is beginning to feel nervous about travelling alone. Hanna opened up the Accomptrainy app she had recently downloaded on a friend's recommendation to create a trip home from UTS.

Fig D.2. User Experience Story 2: Joining a Group

Hanna is exhausted, and anxious about travelling at night. Her anxiety turns to fear as one of the group members becomes threatening.
It's 12pm on a Wednesday night, and Hanna had a friend recommend the app Accomptrainy, where she can find groups of people to travel on the train with. Given that she doesn't have any readily available friends having had an exhausting day of class, she opts to join an already-created group to travel with instead of making her own. However, she finds that a group member keeps asking inappropriate questions and finds his presence threatening. Hanna pressed the inbuilt emergency button in Accomptrainy.

Appendix E. Scenarios

Fig E.1. Scenario 1

Create a trip that leaves UTS at 8:30pm and is heading to Hurstville. Accept an application from the various users that have applied to the group and invite one of your buddies to the trip as well.

Fig E.2. Scenario 2

Views the available groups on Accomptrainy and selects the first one, which departs from UTS at 8:30 PM and is heading to Hurstville. After inspecting the group members, join them for the trip. During the journey, an incident makes you feel unsafe, navigate to the app's help button to contact emergency assistance.

Fig E.3. Scenario 3

In your most recent past trip an incident occurred during the trip where one of your trip mates, Johnny made you feel uncomfortable. Navigate to your past trips and report Johnny for his actions.

Appendix F. Heuristic Evaluation Results (all results)

Heuristic Evaluation Results (all results)

Appendix G. Recordings of Usability Testing

Recordings of Usability Testing

Appendix H. Usability Evaluation Results

Usability Evaluation Results

Appendix J. Experience Evaluation Recording

Experience Evaluation Recording

Appendix K. Experience Evaluation Results

Experience Evaluation Results